

Sanjeyan Chrysharnthan

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EDUCATION

Nanyang Technological University (NTU), Singapore

Aug 2024 - May 2028

Bachelor of Engineering (Honours), Computer Engineering with Minor in Mathematics

Relevant Coursework: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Object-Oriented Programming

WORK EXPERIENCE

Vertex Holdings, VC Subsidiary of Temasek Holdings

Jul 2026 - Present

Data Scientist Intern | Python, Django, PostgreSQL, GCP, Next.js, pytest

- Enabled natural-language meeting-note queries through an LLM-driven Django engine in a GCP-backed platform; strengthened quality assurance with schema-aware, access-controlled retrieval and 15 tests, passed 3,422 tests, and earned manager praise for cleaner production table output.
- Diagnosed a silent PDF-generation bug clipping investor-facing content; traced the live rendering path and a second crash it exposed, then shipped a mutation-verified regression test.
- Delivered 10 reviewed Django and Next.js changes through daily stand-ups and weekly Agile sprints; traced a CI-ordering defect to process-global factory state, landed a boundary-invariant fix, and passed 2,030 tests.

Panasonic R&D Centre Singapore

May 2026 - Jul 2026

Machine Learning & Computer Vision Engineer (Intern) | Python, PyTorch, FastAPI, Docker

- Built an interactive AI video-analytics platform on Meta's SAM3 that Panasonic R&D Centre Singapore selected as one of four AI technologies for AIMX Singapore x TechInnovation 2026, focused on deployment and commercialization partnerships.
- Built and deployed a FastAPI/React video-segmentation app through Docker on a 24GB GPU server, supporting multi-object point, brush, text, and automatic prompting, mask propagation, and live SSE previews.
- Hardened SAM3 GPU serving by fixing bf16 inference, object-ID collisions, and race conditions; added mocked-GPU tests and MLflow, Prometheus, and Grafana observability.

Rohde & Schwarz Asia

May 2025 - Jul 2025

Software Development Engineer (Intern) | Python, Pandas, NumPy, Git

- Partnered with a supervisor and RF engineer to translate a manual CSV-comparison workflow into an adopted Python/Pandas data quality pipeline; reconciled inconsistent schemas, flagged missing or duplicate records, and auto-populated Excel reports.

PROJECTS & OPEN SOURCE

Algorithmic Trading Backtester | Python, Pandas, NumPy, SciPy, SQLite, FastAPI

Mar 2026 - Present

- Built an event-driven Python backtester with next-bar MARKET/LIMIT/STOP execution, persistent order lifecycles, integrated sizing, short-margin and borrow-cost accounting, walk-forward analysis, and finite-sample-corrected permutation tests.
- Engineered quality-gated yfinance/Alpha Vantage ingestion with exchange-calendar validation, immutable Parquet generations, and lineage; added 11 SQL analytics plus API, CLI, dashboard, and report surfaces, validated by 712 tests, Ruff, and strict typing across 64 modules.

Market Risk Forecasting Lab | Python, PyTorch, statsmodels, DVC, MLflow, FastAPI

Aug 2026 - Present

- Built a synthetic-only market-risk platform comparing HAR/GARCH, gradient boosting, and a PyTorch LSTM under availability-safe, locked walk-forward evaluation; no empirical forecasting performance is claimed.
- Implemented VaR/Expected Shortfall calibration and backtesting, model diagnostics, and cost-aware experiments through reproducible artifacts; validated 204 tests, Ruff, and strict typing across 74 modules.

FinancePy: Merged Open-Source Contribution | Python, Quantitative Finance, pytest

Aug 2026

- Contributed a merged FinancePy fix for Python 3.10/3.11 SyntaxErrors in bond YTM/OAS pricing code that broke the bonds module; found and fixed a second latent instance beyond the original report.
- Validated the compatibility fix with 61 bond-focused tests and a clean Python 3.11 import, then followed the upstream issue-claiming and contribution workflow through maintainer review and merge.

TECHNICAL SKILLS

Languages: Python, SQL, Java, Go, C++, JavaScript, TypeScript

AI & Quant: PyTorch, scikit-learn, statsmodels, Pandas, NumPy, SciPy, time-series modelling, VaR/Expected Shortfall

Systems & Data: Django, FastAPI, React, Next.js, PostgreSQL, SQLite, Docker

Quality & Tools: pytest, Ruff, mypy, Git, GCP, DVC, MLflow, Prometheus, Grafana